MATRIX OF REVISIONS(Title Defense)

A Risk-Based Two-Factor Student Verification and Event Attendance Management System Using NFC, PIN Authentication, and QR Credential Validation

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Section	Student's Original Work	Panel Recommendations/ Concerns	Action Taken
Research Title	"Hybrid NFC-Enabled Student Verification and Event Attendance Management System with Human-in-the-Loop Validation."	"The previous title was too generic and too focused on NFC only." "Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	The title was revised to "A Risk-Based Two-Factor Student Verification and Event Attendance Management System Using NFC, PIN Authentication, and QR Credential Validation."
Project Context	The original study proposed a Hybrid NFC-Enabled Student Verification and Event Attendance Management System with Human-in-the-Loop Validation, where automated verification was combined with manual visual confirmation by security personnel before access decisions.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	The project was redesigned as a Risk-Based Two-Factor Student Verification and Event Attendance Management System that performs automated authentication through configurable verification modes. Security personnel now monitor flagged transactions instead of approving every authentication request.
Purpose / System Design	The study focused on improving identity verification through NFC-based authentication and attendance recording.	"The system was considered outdated because it only relied on NFC." "Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	The revised study now states that it focuses on the design and development of a Risk-Based Two-Factor Student Verification and Event Attendance Management System Using NFC, PIN Authentication, and QR Credential Validation and explains that it enables NFC identification, PIN authentication as a second verification factor, and QR code validation to confirm the integrity of the physical credential.

Statement of the Problem	The original Statement of the Problem focused on integrating NFC-based "Tap-and-Go" authentication, database matching, Human-in-the-Loop visual verification, automated attendance recording, and evaluating the system in terms of accuracy, processing speed, and usability.	"The panel raised security concerns about what happens if someone picks up or uses another student's NFC ID." "The panel questioned the reliability of using only NFC because NFC UIDs may be cloned or misused."	The Statement of the Problem was revised to focus on the development of a Risk-Based Two-Factor Verification System utilizing NFC, PIN authentication, and QR Credential Validation. New objectives addressing adaptive verification modes, credential integrity validation, automated decision-making, and enhanced security monitoring were incorporated.
Research Objectives	The original objectives emphasized improving campus verification through NFC authentication, enrollment validation, and attendance management.	"The panel recommended a more advanced research direction."	The research objectives were expanded to include the implementation of administrator-configurable Fast Mode, Standard Mode, and High-Security Mode, PIN authentication, QR credential validation, security alert generation, entry/exit monitoring, and restricted-event verification while maintaining efficient attendance management.
Conceptual Framework (IPO)	The original IPO framework illustrated Input → NFC Verification → Human-in-the-Loop Validation → Attendance Recording → Output, where security personnel manually confirmed student identity before access was granted.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	The IPO framework was redesigned to illustrate Input → Risk Assessment → Verification Mode Selection → NFC Authentication → PIN Verification (if required) → QR Credential Validation (High-Security Mode) → Automated Decision → Security Monitoring and Reporting. Human intervention was shifted from routine verification to monitoring flagged transactions and responding to security alerts.

Scope of the Study	The original scope limited the prototype to NFC-based student verification, enrollment validation, and event attendance management with Human-in-the-Loop confirmation.	"The panel noted that the system's impact seemed too small."	The scope was expanded to include Risk-Based Two-Factor Authentication, PIN authentication, QR Credential Validation, administrator-configurable verification modes, restricted-event management, entry/exit tracking, anti-tailgating mechanisms, credential mismatch detection, security alerts, and a centralized WinUI administrative dashboard while maintaining the system as a prototype implementation.
Limitations of the Study	The original study stated that the prototype was limited to NFC-based verification and did not include biometric authentication, QR validation, or other advanced authentication technologies.	"The panel suggested removing or reducing early limitations."	The limitations were revised to clarify that the system implements NFC and PIN as the two authentication factors, with QR Credential Validation used solely to verify credential integrity. The study also specifies that biometric authentication, facial recognition, and hardware durability testing remain outside the scope of the prototype.
Significance of the Study	The original significance highlighted the benefits of NFC technology combined with Human-in-the-Loop validation for improving student verification, attendance recording, and campus security.	"The panel recommended a more advanced research direction."	The Significance of the Study was rewritten to emphasize the contribution of a Risk-Based Two-Factor Authentication Framework integrating NFC, PIN authentication, and QR Credential Validation, improving operational efficiency, campus security, and institutional decision-making.

Human-in-the-Loop Verification	The original system relied on Human-in-the-Loop Validation, where security personnel manually compared the student's physical appearance with the information displayed by the system before allowing entry.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	Human-in-the-Loop authentication was replaced with automated risk-based verification. Security personnel no longer approve every transaction but instead monitor flagged transactions, security alerts, and credential mismatch notifications.
Authentication Architecture	Authentication depended primarily on NFC Tap-and-Go verification, where possession of a valid NFC-enabled student ID was generally sufficient for successful verification after database validation.	"The system was considered outdated because it only relied on NFC."	The authentication architecture was redesigned into a Risk-Based Two-Factor Authentication Framework, combining NFC (Something You Have) and PIN Authentication (Something You Know) while adapting verification requirements according to administrator-selected security modes.
PIN Authentication	The original prototype did not require users to enter a PIN or any additional knowledge-based credential after tapping their NFC ID.	"The panel raised security concerns about what happens if someone picks up or uses another student’s NFC ID."	A PIN Authentication module was introduced as the second authentication factor. Students are required to enter their registered PIN after successful NFC identification, with incorrect attempts recorded for account lockout protection.
QR Credential Validation	The original verification process relied solely on the NFC UID stored in the student ID and did not verify whether the physical credential itself had been altered or cloned.	"The panel questioned the reliability of using only NFC because NFC UIDs may be cloned or misused."	QR Credential Validation was added; during High-Security Mode, the QR code on the ID is compared with the student's registered credential information. Mismatched credentials automatically generate a Credential Mismatch alert and deny access.
Fast Mode	The original system processed every authentication request using a single verification workflow regardless of traffic conditions or security requirements.	"The panel raised concerns about high-traffic entry and practical deployment."	Fast Mode was introduced; this mode performs NFC authentication only, maximizing throughput for routine campus entry while maintaining database validation.

Standard Mode	The original system did not differentiate between different verification levels or authentication requirements.	"The panel raised concerns about high-traffic entry and practical deployment."	Standard Mode was implemented as the default verification mode, requiring NFC authentication followed by PIN verification for stronger identity assurance.
High-Security Mode	The original prototype did not provide additional authentication procedures for high-risk situations, sensitive areas, or restricted university events.	"The panel recommended a more advanced research direction."	High-Security Mode was introduced, performing the complete authentication workflow: NFC authentication, PIN verification, and QR Credential Validation.
Entry/Exit State Tracking	The original prototype primarily recorded student entry and attendance transactions; it did not distinguish movement history.	"The panel suggested monitoring for both entrance and exit."	The system was enhanced with Entry/Exit State Tracking, allowing every successful verification to be classified as ENTRY, EXIT, or EVENT ATTENDANCE.
Anti-Tailgating Mechanism	The original system did not include a mechanism to prevent repeated use of the same credential for unauthorized entry or tailgating incidents.	"The panel raised concerns about outsiders entering through backdoors or with help from insiders."	An Anti-Tailgating Mechanism was introduced by maintaining the current access state; if a student is already inside, another entry attempt is denied until an exit is recorded.
Restricted Event Validation	The original attendance module recorded attendance regardless of whether the student was authorized for that specific event.	"The panel asked what happens if an official memo limits an event to selected attendees only."	Restricted Event Validation was implemented by verifying that the student is registered for the selected event before recording attendance.
Security Alert Generation	The original prototype did not actively notify administrators of suspicious or abnormal verification attempts.	"Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	A Security Alert Module was integrated to automatically notify administrators of credential mismatches, repeated failed PIN attempts, unauthorized event access, and other suspicious activities.
PIN Lockout Protection	The original system contained no protection against repeated authentication attempts.	"The panel raised security concerns about what happens if someone picks up or uses another student's NFC ID."	PIN Lockout Protection was implemented, automatically locking PIN verification after three consecutive incorrect attempts.

Credential Mismatch Detection	The original verification process trusted the NFC credential after successful database validation.	"The panel questioned the reliability of using only NFC because NFC UIDs may be cloned or misused."	Credential Mismatch Detection was introduced via QR Credential Validation to identify altered or duplicated identification cards.
Security Logging and Audit Trail	The original system stored records primarily for reporting purposes, without comprehensive security categorization.	"The panel suggested monitoring for both entrance and exit."	The revised system maintains a comprehensive Security Audit Trail recording successful authentications, failed PIN attempts, credential mismatches, unauthorized event access, and administrator actions.
Self-Service PIN Recovery (OTP)	The original prototype did not provide a mechanism for autonomous credential recovery.	"Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	A decentralized, student-facing web portal was incorporated to allow for out-of-band email OTP PIN resets, ensuring autonomous credential recovery without burdening administrators.
Figure 11 – Operational Flowchart	The original flowchart illustrated a linear verification process relying on Human-in-the-Loop.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	Figure 11 was redesigned into a Risk-Based Operational Flowchart featuring verification modes, PIN verification, QR validation, and automated decision-making.
Figure 13 – Use Case Diagram	The original diagram was centered around Human-in-the-Loop verification.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	Figure 13 was updated to include Enter PIN, Scan QR, Monitor Security Alerts, Respond to Flagged Transactions, and Set/Reset Student PIN.
Figure 18 – Student Verification Workflow	The original workflow was a sequential process ending with manual confirmation.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	Figure 18 was redesigned to incorporate risk assessment, verification mode selection, PIN authentication, QR Credential Validation, and automated verification decisions.
Figure 19 – Event Attendance Workflow	The original workflow focused on recording attendance after successful NFC verification.	"The panel asked what happens if an official memo limits an event to selected attendees only."	Figure 19 now validates event registration, applies risk modes, performs PIN/QR validation, and generates alerts for unauthorized attempts.

Data Flow Diagram (DFD)	The original DFD emphasized manual confirmation and NFC retrieval.	"The panel suggested removing heavy reliance on human-in-the-loop validation."	The DFD was revised to include data flows for risk evaluation, PIN verification, QR Credential Validation, security alert generation, and restricted-event validation.
Entity Relationship Diagram (ERD)	The original ERD stored only student info, NFC credentials, and basic attendance.	"The panel recommended a more advanced research direction."	The ERD was expanded to include PIN credentials, verification modes, restricted-event registration, entry/exit records, credential mismatch logs, and security alerts.
System Architecture	The original architecture relied on NFC and Human-in-the-Loop verification.	"Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	The revised architecture integrates ESP32, PN532 NFC Reader, USB Webcam, WinUI Administrator Dashboard, and a Hybrid MySQL–Firebase Database.
Hybrid Database Architecture	The original architecture synchronized basic NFC and attendance records.	"The panel recommended a more advanced research direction."	The architecture now supports PIN credentials, risk-mode configurations, mismatch records, event registrations, entry/exit transactions, and security alerts.
WinUI Administrator Dashboard	The original interface had limited security monitoring capabilities.	"The panel suggested a kiosk-type monitoring setup."	The dashboard was enhanced to manage verification modes, monitor security alerts, review flagged transactions, configure restricted events, and generate reports.
Hardware Architecture	The original hardware consisted only of the ESP32, NFC reader, and an onboard RTC.	"The panel questioned why NFC should be used instead of QR code or facial recognition."	A USB Webcam was integrated to perform QR Credential Validation, while the ESP32 maintains NFC and keypad handling, replacing the onboard RTC with computer-based timestamping.

Unimplemented or Out-of-Scope Panel Recommendations

Panel Recommendations/ Concerns	Unimplemented Component	Status / Justification for Defense

"Panelists suggested adding advanced methods such as codes, QR, OTP, biometrics, facial recognition, and notifications."	Biometric Authentication & Facial Recognition	Out of Scope (Limitation): Excluded from the current prototype due to higher hardware costs and increased computational overhead on the ESP32 microcontroller, as well as data privacy considerations.
"The panel questioned why NFC should be used instead of QR code or facial recognition."	Complete replacement of NFC infrastructure	Partially Addressed / Defended: NFC was retained as the primary identification mechanism ("Something You Have") because of its durability and speed, but its cloning vulnerability was mitigated by overlaying PIN and QR validation rather than replacing it entirely.